MIL-I-15883B

19 December 1967

SUPERSEDING
MIL-I-15883A
11 September 1963

MILITARY SPECIFICATION

INK, WRITING, PERMANENT, SEAWATER AND FUEL OIL RESISTANT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

* 1. SCOPE

1.1 Scope.- This specification covers a permanent writing ink resistant to seawater and fuel oil.

1.2 Classification.- The ink covered by this specification shall be of the following colors as specified (see 6.2):

Black
Blue

* 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-P-121	-	Paper, Bond and Writing, White and Colored.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-636	-	Box, Fiberboard.
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing.
PPP-T-76	-	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (For Carton Sealing).

STANDARDS

FEDERAL

Fed. Std. No. 595 - Colors.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage:

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 Materials.- Materials used in manufacture of the ink shall conform to the requirements set forth below.

* 3.1.1 Writing qualities.- The ink shall be suitable for use with either fountain or dip-type pen and shall flow freely and evenly without gumming, drying or flaking on the pen point. The writing shall be free of feathered or ragged edges and shall not "strike through" the paper when tested as specified in 4.3.2.

* 3.1.2 Smudge or offset.- The ink shall not smudge or offset onto adjacent sheets of paper when tested as specified in 4.3.2.

* 3.1.3 Stability.- The ink shall not coagulate, deposit surface or bottom sediment and shall be free of mold when tested as specified in 4.3.2.

* 3.1.4 Corrosiveness.- The ink shall not highly etch, corrode, or pit the carbon steel pen point to affect its writing qualities when tested as specified in 4.3.2.

* 3.1.5 Color.- The color shall conform to Black No. X37038 and Blue No. X35177 of Fed. Std. No. 595, when tested as specified in 4.3.2.

* 3.1.6 Permanency.- Written script produced with the inks shall be legible after submersion in seawater and fuel oil for 28 days when tested as specified in 4.3.2.

* 3.2 Workmanship.- The finished product shall be free from foreign matter.

* 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to

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perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105 except where otherwise indicated hereinafter.

4.2.1 Inspection of materials and components.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended or qualified in this specification or applicable purchase documents.

4.3 Inspection of the end item.-

4.3.1 Examination of the end item.- The end item shall be examined for the defects in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQLs) set forth in 4.3.1.4. The lot size shall be expressed in units of filled unit containers of the specified capacity for the examinations in 4.3.1.1 and 4.3.1.2 and in units of shipping containers for the examination in 4.3.1.3.

4.3.1.1 Examination of the end item for visual defects.- The sample unit for this examination shall be one filled unit container.

Examine	Defect
Container	Not size, shape, type container specified. Any evidence of seepage or leakage from any part of container including closure.
Contents	Not color required (see 6.2). Presence of foreign matter.
Markings	Missing, incomplete, incorrect, not as specified (see 5.3.1).

4.3.1.2 Examination for average net contents.- The sample unit for this examination shall be one filled unit container. The lot shall be unacceptable if the average net contents per container for all sample units examined is less than specified or indicated.

4.3.1.3 Examination of preparation for delivery.- An examination shall be made to determine that packaging, packing and marking requirements comply with the requirements of Section 5 of this specification. The sample unit shall be one shipping container, fully prepared for delivery except that it need not be sealed. The lot shall be the number of shipping containers in the end item inspection lot. Closed shipping containers shall be examined for closure defects.

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Examine	Defect
Markings	Incorrect, incomplete, illegible, omitted; of improper size, sequence or method of application; not in accordance with contract requirements.
Material	Any nonconforming component; component missing, damaged or otherwise defective.
Workmanship	Inadequate application of components such as: Incomplete closure of container flaps, loose or inadequate sealing, strapping or stapling. Bulged or distorted container.
Contents	Number of primary containers per shipping container is less than specified.

4.3.1.4 Inspection levels and acceptable quality levels (AQLs) for examination.- The inspection levels for determining the sample size and acceptable quality levels (AQLs), expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection level	AQL
4.3.1.1	I	2.5
4.3.1.2	S-2	N/A
4.3.1.3	S-2	2.5

4.3.2 Testing of the end item.- The end item shall be tested for the characteristics specified in table I. The lot size shall be expressed in pints. The sample unit for testing shall be one pint composite. The composite shall be made by thoroughly mixing equal portions selected at random throughout the lot in the frequency listed below. The composite shall be placed in a clean, dry glass container. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if the composite fails to meet any test requirement specified.

Lot size (pints)	Number of portions to form composite
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

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Table I Instructions for testing of the end item

Characteristic	Specification reference		Number Determinations per composite unit	Results reported as Pass or Fail
	Requirement	Test method		
Writing qualities	3.1.1	4.4.1	1	X
Smudge or offset	3.1.2	4.4.2	1	X
Stability	3.1.3	4.4.3	1	X
Corrosiveness	3.1.4	4.4.4	1	X
Color	3.1.5	4.4.5	1	X
Permanency				
Seawater	3.1.6	4.4.6	1	X
Fuel oil	3.1.6	4.4.7	1	X

4.4 Test procedures.-

* 4.4.1 Writing qualities and preparation of test sheets.- In determining the writing qualities fountain and dip-type pens equipped with clean, smooth medium points shall be used. The first sentence of this paragraph shall be written with each pen on each of 16 sheets (8 x 5-1/4 inches) of white bond paper conforming to UU-P-121, type III (substance 40). Each pen shall write when the point first touches the paper without undue pressure to start the flow of ink. The ink shall flow smoothly and uniformly without skipping, dripping or blotting. The writing shall be free of feathered or ragged edges and shall not "strike through" the paper. The filled fountain pen shall be placed in a horizontal position for 5 minutes with the cap removed. The first sentence of this paragraph shall be rewritten on the same sheets of paper. The ink shall flow freely and evenly without gumming, drying or flaking on the pen point. The written sheets shall be air dried for three hours and used as specimens for tests in 4.4.6 and 4.4.7.

* 4.4.2 Offset test.- Simulate signing letters with a fountain or dip pen with medium point by writing a signature on five successive sheets of paper under atmospheric conditions of 50 $\pm$ 2 percent relative humidity and a temperature of 73° $\pm$ 3.5°F. Allow each sheet to fall on top of the previous sheet five seconds after the signature is finished. The writing shall not smudge or offset onto the adjacent sheet of paper.

* 4.4.3 Stability.- Place 100 milliliters of ink in a beaker and cover the beaker with a watch glass to exclude dust, yet permitting access of air. The beaker shall be conditioned in a room maintained at a temperature of 73°F

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(± 3.5°F) and a relative humidity of 45-55 percent for 14 days. The sample shall show no formation, coagulation, surface or bottom sediment when filtered through a Whatman No. 1 filter paper or equivalent.

* 4.4.4 Corrosiveness.- Immerse a common carbon steel pen point by a cotton thread in no less than four ounces of ink for one week. At the end of the week, the pen point shall be in such serviceable condition that it may be used for writing. The pen point may be slightly etched, corroded or pitted.

* 4.4.5 Color.- Apply the ink with pen or fine brush to bond writing paper conforming to UU-P-121, type III (substance 40) to obtain maximum darkness. Compare with applicable color chip in Fed. Std. No. 595.

* 4.4.6 Resistance to seawater.- Immerse four test sheets prepared in 4.4.1 in synthetic seawater, maintained at a temperature of 73 ± 5°F. The seawater shall have the following composition:

Magnesium chloride	$MgCl_2.6H_2O$	11.0 gram
Calcium chloride	$CaCl_2.2H_2O$	1.6 gram
Anhydrous sodium sulphate	Na_2SO_4	4.0 gram
Sodium chloride, C. P.	NaCl	25.0 gram
Water	H_2O to make	1 liter

At the end of each seven-day period remove one sheet, rinse thoroughly with distilled water and air dry. Examine the writing on each sheet for legibility.

* 4.7 Resistance to fuel oil.- Immerse 12 test sheets prepared in 4.4.1 in No. 2 grade fuel oil maintained at a temperature of 73 ± 5°F. At the end of each seven-day period remove three test sheets from the oil, remove excess oil with a rubber squeege. Examine the writing on each sheet for legibility.

* 5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.-

* 5.1.1.1 Unit packaging.- Ink shall be furnished in 4-ounce or 1-pint commercial type bottles as specified (see 6.2). The bottles shall be securely closed with non-corrosive metal caps. Each 4-ounce bottle shall be individually packaged in a snug-fitting paperboard box conforming to variety 1, style II, type A, class a of PPP-B-566, and each 1-pint bottle shall be packaged in a snug-fitting corrugated fiberboard box conforming to style TS, type CF, class domestic, variety SW, grade 125 of PPP-B-636. Both ends of the box shall be securely closed with minimum 2-inch width tape conforming to PPP-T-76, or type III, grade B of PPP-T-45.

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* 5.1.1.2 Intermediate packaging.- Twelve 4-ounce bottles of one color ink only, packaged as specified in 5.1.1.1, shall be placed in a snug-fitting corrugated fiberboard box conforming to style RSC, type CF, class domestic, variety SW, grade 200 of PPP-B-636. Each box shall be closed by covering all seams with minimum 2-inch width tape conforming to PPP-T-76, or type III, grade B of PPP-T-45. Intermediate packaging shall not be required for 1-pint bottles of ink.

* 5.1.2 Level C.- Bottled ink shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

* 5.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

* 5.2.1 Level A.- Forty-eight 4-ounce bottles, or twelve 1-pint bottles, of one color ink only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Closure, waterproofing by means of tape, and reinforcing with strapping or tape banding shall be in accordance with the appendix of PPP-B-636.

* 5.2.2 Level B.- Forty-eight 4-ounce bottles or twelve 1-pint bottles, of one color ink only, packaged as specified in 5.1, shall be packed in a snug-fitting corrugated fiberboard shipping container conforming to style RSC, type CF, class domestic, variety SW, grade 275 of PPP-B-636. Closure shall be in accordance with method II as specified in the appendix of PPP-B-636.

* 5.2.2.1 When specified (see 6.2), the shipping container shall be grade V3c or V3s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix thereto.

* 5.2.3 Level C.- Bottled ink, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.-

5.3.1 Label.- The following legend shall be legibly printed in black on a white label, and affixed to each bottle:

Federal Stock No.
Nomenclature and color of ink.
Manufacturer's name and location.
Volume of contents.

* 5.3.2 Boxes and containers.- In addition to any special marking required by the contract or order, interior boxes and shipping containers shall be marked in accordance with MIL-STD-129. In addition, the top of each shipping container containing glass bottles shall be marked:

"GLASS - THIS SIDE UP".

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6. NOTES

6.1 Intended use.- The ink covered by this specification is intended for use for permanent handwritten records which inadvertently may come in contact with seawater or fuel oil.

* 6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Color required (see 1.2).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
d. Package size (see 5.1.1.1).
e. When weather-resistant grade V3c or V3s fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from prior issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Air Force - 84

Review activities:

Army - GL, MD
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User activities:

Navy - SA, MC

Preparing activity:

Army - GL

Project No. 7510-0161

Review and user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Documents.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION: MIL-I-15883B - Ink, Writing, Permanent, Seawater and Fuel Oil Resistant

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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